

Zhe Chen 陈哲

zc2325@ic.ac.uk / ssyzc1011@gmail.com

github.com/CZ114

Personal website: <https://cz114.github.io/>



Education

Imperial College London

2025.10 – 2026.09

MRes, Medical Robotics | Research completed; formal degree award pending

University of Nottingham Ningbo China

2021.09 – 2025.06

BEng (Hons), Electrical and Electronic Engineering | **First-Class Honours**

Research Experience

Imperial College London

Hamlyn Centre · I-thread Lab | MRes Research / Research Assistant (current)

Multimodal AI Analysis Platform for Alzheimer's Disease Research

2026.01 – 2026.09

MRes Research

Poster exhibited · Hamlyn Symposium on Medical Robotics 2026

Developed a **wearable-to-agent research platform** for multimodal monitoring, integrating ESP32 sensing, signal processing, retrieval and agent workflows into evidence-linked reports.

- **Multimodal sensing:** Integrated PPG, GSR, IMU and audio on ESP32-S3; sustained approximately 50 Hz acquisition with no detected frame loss across **136,562 frames / 27 serial recordings**.
- **Heart-rate estimation:** Combined spectral analysis, autocorrelation and motion suppression; achieved MAE 5.57 bpm vs 11.84 bpm for the device baseline (PPG-DaLiA, 15 participants, LOSO).
- **Agent workflows:** Built deterministic tools, retrieval, memory and bounded workflows; improved movement accuracy from 35.7% to 81.4% in a fixed-model ablation, with parsing failures reduced from 15 to 0.
- **Movement recognition:** Used 18-feature inputs for 5-class recognition; achieved **82.9% accuracy / 0.804 macro-F1**, versus 78.6% / 0.713 for random forest (70 recordings, 6 healthy volunteers, LOSO).
- **RAG and retrieval:** Implemented hybrid retrieval and HyDE comparisons; achieved Recall@10 = 0.846 and MRR = 0.753–0.774 on a 24-question internal set.
- **Tool-use and reporting:** Aligned the in-house BFCL scorer with official decisions on **400/400 cases** (both scored 91.25%); **66/70 (94.3%)** reports met all traceability checks.
- **AgentClinic simulation:** Reduced mean dialogue turns for all six models; two model conditions improved task accuracy from 53% / 77% to 67% / 87%.

Speech Evidence & Clinician Report Chat

2026

Research Assistant | Collaborative research support

Manuscript in preparation · Target journal: Nature Medicine

- Assisted with checking and verifying speech metrics within the collaborative research project.
- Helped organise the report page and existing content for review.
- Implemented part of the chat functionality within the report interface.

Research Experience (continued)

University of Nottingham Ningbo China (UNNC)

Research, undergraduate dissertation and embedded systems

Smart-Home IoT and Federated Learning for Sleep Apnoea

2023.09 – 2025.07

UNNC Smart Medicine Laboratory | Research Intern / Project Lead

- Built an M5Stack/ESP32 + MQTT gateway and FATE/PyTorch experiments for home heart-rate data without expert labels. Ran confidence-thresholded pseudo-labelling and four federated training rounds, comparing pseudo-labels with a control treating home samples as normal.
- Achieved **81% external-test accuracy** in round four versus 76% for the control (+5 percentage points); sleep-apnoea class precision / recall were 0.93 / 0.78.
- Contributed experimental evidence and manuscript drafting as a listed co-author on collaborative modelling with unlabelled home data.

Reinforcement Learning for Port Container Relocation

2024.08 – 2025.06

Undergraduate Dissertation

- Built a three-dimensional container-yard simulator to reduce unnecessary handling; designed state representations and reward functions, and implemented A2C/DQN training and evaluation pipelines.
- Against a random-action baseline, DQN reduced rehandles by approximately 23% in simple scenarios, while A2C reduced handling moves by approximately **40% in complex scenarios**.

DeskPal – Embedded AI Companion

2024.12

Embedded ML and IoT | Coursework (EEEE3124)

Best IoT Project Award • UNNC

- Built a Wio Terminal desktop companion; trained an Edge Impulse MLP for three motion classes (idle, rise and wave) and deployed on-device inference.
- Integrated environmental sensing, LCD expression / reward states and bidirectional MQTT for interactive feedback, reminders and data exchange.

Technical Skills

Agents and Evaluation: Function Calling, RAG, workflows and memory; BFCL calibration, retrieval evaluation, LOSO, ablation studies and evidence traceability.

Algorithm Foundations: Python, PyTorch, A2C/DQN, federated learning and pseudo-labelling; systematic study of Transformer/Attention and the principles of SFT, RLHF and PPO.

Robotics and Embedded Systems: ESP32-S3, I2S, BLE, UDP, MQTT, CoppeliaSim, robot kinematics, Ubuntu, Jetson Nano and Altium Designer.

Open-Source Tools

Vibedog-for-agents 2026.05 – 2026.06 | AI-assisted development

Agent activity display, human approval and a stop control. github.com/CZ114/Vibedog-for-agents

esp32-audio-ai | Audio acquisition and transcription precursor to the main project

ESP32 audio capture, Whisper transcription, and pause / speech-rate statistics.

github.com/CZ114/esp32-audio-ai